Problem 7

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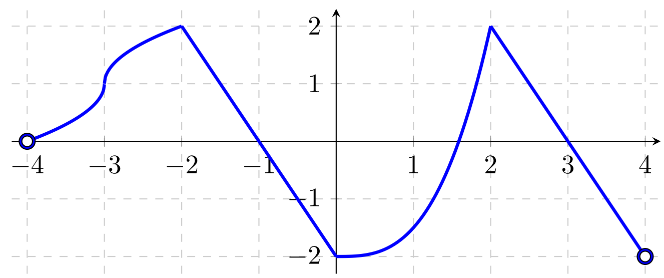
Problem 1 *The graph of a function $g(x)$ with domain $(-4, 4)$ is given in the figure. This portions of this graph in the intervals $(-2, 0)$ and $(2, 4)$ are straight lines. The portion on the interval $(0, 2)$ is not parabolic.*



For each of the limits below, give the form of the limit, then evaluate the limit.

(a) $\displaystyle\lim_{x\to -3} \frac{g(x) + \sin\left(\frac{\pi}{4}x\right)}{x\sqrt{x+4}}$.

(b) $\displaystyle\lim_{x\to 0^+} \frac{e^x}{|g(x)+2|}$.

(c) $\displaystyle\lim_{x\to 3} \frac{x^2+2x-15}{x+g(x)-3}$